

of each menu is explained below.

Designer. Write project name (say, project) and then click 'New/Open' button. A new project directory named 'project' is created in the bin directory. All the files associated with the new project will be stored in the newly created directory (project).

Now we have to generate different LED patterns. We begin with a classic forward chaser (refer Fig. 7). Each new

LED pattern is called 'group' and we need to add these groups to the new project. Let us name the first group as 'forward' (refer Fig. 6). Type 'forward' under 'Group Manager' and click 'New Group.' A newly created group named 'forward' is listed in the table just below the LED buttons. This group needs to be defined using the designer tools such as LED, push and complement buttons and delay. The delay between each step, say,

400 millisecond (very small delay may not be caught by human eye, even though designer allows delay as small as 1ms) has to be defined.

LED buttons.

These buttons are used to turn on/off the LED. LED is turned-on once the button is clicked and turned-off in next click. There are eight LED buttons. These simply state the cur-

rent LED pattern.

Push button. When this button is clicked, current LED pattern is pushed to the table. This means that the table is ready to accept next pattern (or step).

'>' button. Right-shifts current pattern by one step.

'<' button. Left-shifts current pattern by one step.

Compliment. Each turned on LED gets off and each turned off LED gets on.

Table. LED patterns are stored in this table one by one. The LED pattern for 'forward' is shown in Fig. 7.

Enter the eight patterns using LED buttons. After entering the pattern, check the table to verify data. The data in this table can be manually edited. If the data in table is as desired, click the 'Done' button.

Similarly, we make two (or more if you desire) groups named 'reverse' and 'dancing' (refer Figs 8 and 9). 'Done' button is pressed after designing each group.

Organiser. Use the tab 'Organiser' to organise your project (refer Fig. 10). In our project there are three groups—'forward', 'reverse' and 'dancing'—in the group named 'project.' A group can be repeated several times in a project. For example, to repeat the group 'forward' 20 times in this project, select 'forward' from list and put 'repetition=20' and click '=' button. The table gets updated. Similarly, update other groups until you get the table like this:

```
forward—20
dancing—10
reverse—20
dancing—10
```

The table represents that the 'forward' pattern is executed 20 times, then 'dancing' 10 times, followed by 'reverse' 20 times and 'dancing' again 10 times. The whole process is repeated until stopped. Click 'Done' after completion.

Simulator. Once the LED groups are designed and organised, the LED pattern needs to be verified. Each

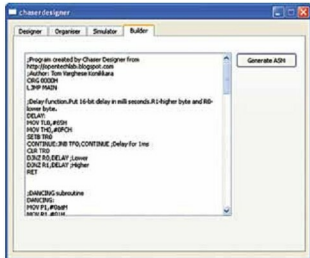


Fig. 12: Builder windows

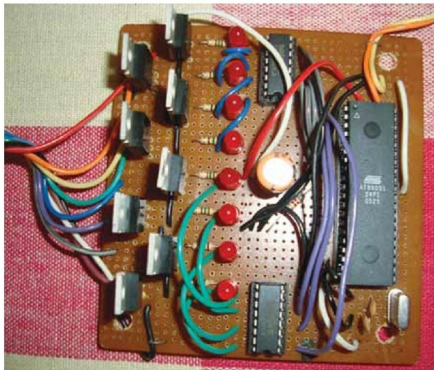


Fig. 13: Author's prototype